Northeastern University

MATH 6243: Statistical Learning

Spring 2026 Course Syllabus (CRN: 38869)

Professor Info

Instructor:	Dr. Eric Gerber	**Office:**	ME 331
Email:	e.gerber@northeastern.edu	**Email Prefix:**	[MATH 6243]:
Canvas Page:	Canvas Login		
Gradescope:	Gradescope Link	**Piazza:**	Piazza Link

Class Meeting Times/Classrooms:

- TR 5:50 pm – 7:30 pm (**Richards Hall 233)**

Prerequisites:

- MATH 5010 (with a minimum grade of C-)

Description:
Presents fundamental principles of machine learning from a statistical standpoint. Focuses on supervised learning, including regression and classification methods. Topics include linear and polynomial regression, logistic regression, and linear discriminant analysis; cross-validation and the bootstrap, model selection, and regularization methods (ridge and lasso); nonlinear models, splines, and generalized additive models; tree-based methods, random forests, and boosting; and support-vector machines. Examines methods used to learn from data taking a deeper look at algorithms that may appear to be different but actually use similar principles. Discusses some unsupervised learning methods principal components and clustering. Uses a programming language to fit these models using modern techniques for visualization and reporting.

Tentative Course Outline:

- Week 1: Introduction, Python, Supervised ML
- Week 2: Regression (Linear, Polynomial)
- Week 3: Regression (Regularization)
- Week 4: Classification (Logistic Regression)
- Week 5: Classification (LDA)
- Week 6: Bagging and Boosting
- Week 7: Ensemble Methods (Random Forests)
- Week 8: Smoothers and Loess
- Week 9: Generalized Additive Models
- Week 10: GAMs and Bayesian Analysis
- Week 11: Bayesian Analysis
- Week 12: SVMs, PCA
- Week 13: Clustering, Intro to NNets
- Week 14: Deep Learning

Communication: Please check your email, Piazza, and Canvas regularly. **Be aware that attendance is expected**, even if it is not taken; you cannot expect to master the material without attending class. **Piazza** should be your first recourse for asking questions if you cannot attend office hours. **Extreme situations** may arise throughout the semester and permit the professor to allow late homework/lab submission or allow exam retakes. You are not expected to provide complete details, but **must** communicate with the professor **at latest within 24 hours after** of having missed a deadline to discuss an extension. There will be no exceptions.

Office Hours: The professor is happy to meet students in his office, **Meserve Hall 331**, on a first-come, first-serve basis. Given a somewhat odd schedule with meetings this semester, the office hours are planned for:

- **(Always)** Monday and Thursday 11:45 am – 1:15 pm
- **(At Least One of)** Tues 4:00 pm – 5:30 pm, Thurs 4:00 pm – 5:30 pm **or** Wednes 11:45 am – 1:15 pm
  - You can see Dr. Gerber's weekly office hours schedule **here**.

I am often in the office outside these hours and you are always welcome to stop by and see if I have time. You may also **always** email me, or ask after class, to set up a time for us to meet that works best for you.

Materials: This class involves both working out math problems by hand and writing and running computer code. Each student should bring their laptop, pen and paper to each class.

Textbook and References:

- *Elements of Statistical Learning* (primary textbook).
  https://hastie.su.domains/ElemStatLearn/
- *An Introduction to Statistical Learning* (lower level with python specific examples).
  https://www.statlearning.com/
- *Hands-On Machine Learning with Scikit-Learn, Keras and TensorFlow* (more python).
  https://github.com/ageron/handson-ml3
- *Bayesian Data Analysis* (for Bayesian ML).
  http://www.stat.columbia.edu/~gelman/book/

Lectures: Students are expected to behave in a manner that fosters the learning of all students.

Homework (24%): Homework will typically be assigned and due every week or every other week via Gradescope. The lowest HW score will be dropped. Please also refer to the **Academic Integrity** policy on the next page.

Labs (12%): There are several planned lab assignments. They will be primarily be done in-class, but may occasionally (if necessary) be take home assignments. Labs are low-stakes assessments meant to give students hands-on practice. They are graded for completeness rather than for correctness. Please also refer to the **Academic Integrity** policy on the next page.

Project (24%): There will be an open-ended final project allowing students to work in pairs on a novel real data problem of their choice. The project will consist of two phases and four total deliverables: Phase 1 (project pitch and proposal: 2% each) will be due after Spring Break. Phase 2 (project poster and report: 10% each) will be due at the end of the semester. More details will be provided in the project guidelines.

Midterm Exams (20% each): There will be two midterm exams covering the first third and second third of the material and a cumulative final exam during finals week.

Total Grading Policy: HW (24%), Labs (12%), Project (24%), Midterms (20% each).

Important Tentative Dates:

Midterm 1	February 12
Spring Break (no class)	March 2-5
Project Phase 1 due	March 10
Midterm 2	March 26
Project Phase 2 due	April 16

Academic Integrity: Under no circumstances may one student view or share their ungraded homework or exam with another student. This does not extend to discussion of concepts or ideas, but prohibits any sharing of final answers to graded problems.

- please also see Canvas for the student guide on for working with AI tools.
  - any response to any problem on any assignment that is found to have been generated by AI (or copied from any other source) will result in a zero for the entire assignment and the student will be immediately reported to OSCCR: http://www.northeastern.edu/osccr/academic-integrity.
  - Repeat offenses will result in failure of the course.

Disability Access Services (DAS): The office is available to assist students who have a legally documented disability or students who suspect that they may have a disability. If you have a disabling condition that may interfere with your ability to successfully complete this course, please contact Disability Access Services: https://disabilityaccessservices.sites.northeastern.edu/.

Student Resources: Your health and wellness is more important than any assignment. Please use the resources below if you are struggling, and don't hesitate to ask for my help!

- **Counseling Center (24/7 support): 877-233-9477** or https://www.northeastern.edu/uhcs/

Title IX: I am a mandatory reporter under Title IX, which means that I am required to report any and all allegations of discrimination to the Title IX coordinator.

Title IX of the Education Amendments of 1972 protects individuals from sex or gender-based discrimination, including discrimination based on gender-identity, in educational programs and activities that receive federal financial assistance.

Northeastern's Title IX Policy prohibits Prohibited Offenses, which are defined as sexual harassment, sexual assault, relationship or domestic violence, and stalking. The Title IX Policy applies to the entire community, including male, female, transgender students, faculty and staff.

If you or someone you know has been a survivor of a Prohibited Offense, confidential support and guidance can be found through University Health and Counseling Services staff and the Center for Spiritual Dialogue and Service clergy members. By law, those employees are not required to report allegations of sex or gender-based discrimination to the University.

Alleged violations can be reported non-confidentially to the Title IX Coordinator within The Office for Gender Equity and Compliance at and/or through NUPD (Emergency 617-373-3333; Non-Emergency 617-373-2121). Reporting Prohibited Offenses to NUPD does **NOT** commit the victim/affected party to future legal action.

Changes to Syllabus: This course syllabus is intended as a guide. The instructor reserves the right to revise any part of the syllabus during the course. Any changes will be announced well in advance, during class time and/or via email.